Chapter 3 • Sets and Functions

Solved Review Exercise 3

MCQs

1. The set-builder form of {1, $\frac{1}{3}$, $\frac{1}{5}$, $\frac{1}{7}$, …} is:

a) $\{x \mid x=\frac{1}{n}, n\in W\}$

b) $\{x \mid x=\frac{1}{2n+1}, n\in W\}$

c) $\{x \mid x=\frac{1}{n+1}, n\in W\}$

d) $\{x \mid x=2n+1, n\in W\}$

Ans: b

The denominators are the odd numbers 2n+1 for n=0,1,2,….

2. If A = { }, then P(A) is:

a) { }

b) {1}

c) {{ }}

d) ∅

Ans: c

The power set of ∅ is {∅}, a set containing the empty set.

3. If U={1,2,3,4,5}, A={1,2,3}, B={3,4,5}, then U−(A∩B) is:

a) {1,2,4,5}

b) {2,3}

c) {1,3,4,5}

d) {1,2,3}

Ans: a

A∩B={3}, so remove 3 from U.

4. For overlapping A and B, n(A−B) equals:

a) n(A)

b) n(B)

c) A∩B

d) n(A)−n(A∩B)

Ans: d

Remove the common part from A.

5. If A⊆B and B−A≠∅, then n(B−A) equals:

a) 0

b) n(B)

c) n(A)

d) n(B)−n(A)

Ans: d

B−A contains the elements of B outside its subset A.

6. If n(A∪B)=50, n(A)=30 and n(B)=35, then n(A∩B)=

a) 23

b) 15

c) 9

d) 40

Ans: b

30+35−50=15.

7. If A={1,2,3,4} and B={x,y,z}, then A×B contains exactly:

a) 13

b) 12

c) 10

d) 6

Ans: b

|A×B|=4×3=12.

8. If $f(x)=x^2-3x+2$, then f(a+1)=

a) a+1

b) a^2+1

c) a^2+2a+1

d) a^2-a

Ans: d

$(a+1)^2-3(a+1)+2=a^2-a$.

9. Given f(x)=3x+1 and f(x)=28, x=

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a) 9
b) 27
c) 3
d) 18
Ans: a
3x+1=28 gives x=9.

10. For f={(1,a),(2,b),(3,b)} from {1,2,3} to {a,b}, which statement is true?
a) f is injective
b) f is surjective
c) f is bijective
d) f is into only
Ans: b
Both codomain elements are images, so f is onto; it is not one-to-one.

2. Write each set in tabular form.

(i) {x|x = 2n,n∈N}: substitute n = 1,2,3,... to get {2,4,6,8,10,...}.
(ii) {x|x = 2m+1,m∈N}: substitute m = 1,2,3,... to get {3,5,7,9,11,...}.
(iii) {x|x=11n,n∈W,n≤10}: {0,11,22,33,44,55,66,77,88,99,110}.
(iv) {x|x is even and 4<x<6}: no even integer lies strictly between 4 and 6, so ∅.
(v) {x|x is odd and 5<x<7}: no odd integer lies strictly between 5 and 7, so ∅.
(vi) {x|x∈Q and $x^2=2$}: $x=\pm\sqrt{2}$ are irrational, so ∅.
(vii) {x|x∈Q and x=−x}: 2x=0 ⇒ x=0, giving {0}.
(viii) {x|x∈R and x∉Q′}: since Q′ is the irrationals, x must be rational, giving Q.

3. For U={1,…,10}, A={2,4,6,8,10}, B={1,2,3,4,5}, C={1,3,5,7,9}, list the requested sets.

(i) A′={1,3,5,7,9}
(ii) B′={6,7,8,9,10}.
(iii) A∪B={1,2,3,4,5,6,8,10}
(iv) A−B={6,8,10}.
(v) A∩C=∅
(vi) A′∪C′=U because A′=C and C′=A.
(vii) A′∪C={1,3,5,7,9}
(viii) U′=∅.

4. Find the single set equal to each expression.

(i) A′ = U−A
(ii) A∩U = A
(iii) A∪U = U
(iv) A∪∅ = A
(v) ∅∩∅ = ∅.

5. Use Venn diagrams or set algebra to verify the printed statements.

(i) The printed A−B = A∪B′ is not an identity. Correct: A−B = A∩B′, because x∈A and x∉B. Counterexample to the printed union form: U={1,2,3}, A={1,2}, B={2}. Then A−B={1}, but A∪B′={1,3}.
corrected to A−B=A∩B′
(ii) A−B⊆B′, so B⊆(A−B)′. Therefore (A−B)′∩B=B.
verified.

6(a). Verify all four properties for A={1,2,3,4}, B={3,4,5,6,7,8}, C={5,6,7,9,10}.

Associative union: (A∪B)∪C = A∪(B∪C) = {1,2,3,4,5,6,7,8,9,10}.
Associative intersection: (A∩B)∩C = A∩(B∩C) = ∅.
Union over intersection: A∪(B∩C) = (A∪B)∩(A∪C) = {1,2,3,4,5,6,7}.
Intersection over union: A∩(B∪C) = (A∩B)∪(A∩C) = {3,4}.
All four properties verified.

6(b). Verify the same properties for A=∅, B={0}, C={0,1,2}.

Associative union: both sides = {0,1,2}; associative intersection: both sides = ∅.
A∪(B∩C) = {0} and (A∪B)∩(A∪C) = {0}.
A∩(B∪C) = ∅ and (A∩B)∪(A∩C) = ∅.

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All four properties verified.

6(c). Verify the properties for A=N, B=Z, C=Q.
Since N⊆Z⊆Q: (N∪Z)∪Q = N∪(Z∪Q) = Q; (N∩Z)∩Q = N∩(Z∩Q) = N.
N∪(Z∩Q) = (N∪Z)∩(N∪Q)
= Z.
N∩(Z∪Q) = (N∩Z)∪(N∩Q)
= N.
All four properties verified.

7. Verify De Morgan's laws for U={1,…,20}, A=evens and B=odds.
A∪B=U and A′=B, B′=A. Hence (A∪B)′=∅=B∩A=A′∩B′.
A∩B=∅, so (A∩B)′=U. Also A′∪B′=B∪A=U.
Both De Morgan laws verified.

8. P={5m | m∈N} and Q={2m | m∈N}. Find P∩Q.
A number divisible by both 5 and 2 is divisible by lcm(5,2)=10.
P∩Q={10,20,30,40,50,…}.

9. Deduce the two displayed results from set properties.
(i) A∩(A∪B)=A by absorption, and A∪(A∩B)=A by absorption; hence they are equal.
(ii) A∪(A∩B)=A and A∩(A∪B)=A; hence they are equal.

10. For g(x)=7x−2 and s(x)=8x²−3, evaluate the six expressions.
(i) g(0)=−2
(ii) g(−1)=−9
(iii) $g(-5/3)=\frac{-41}{3}$
(iv) s(1)=5
(v) s(−9)=645
(vi) s(7/2)=95.

11. f(x)=ax+b, f(−2)=3 and f(4)=10. Find a and b.
−2a+b=3 and 4a+b=10. Subtracting gives 6a=7, so $a=\frac{7}{6}$
b = 3+2a
$= 3+\frac{7}{3}$
$= \frac{16}{3}$
$a=\frac{7}{6}$, $b=\frac{16}{3}$

12. For k(x)=7x−5, find x when k(x)=100.
7x−5=100
⇒ 7x=105
⇒ x=15.

13. g(x)=mx²+n, g(4)=20 and g(0)=5. Find m and n.
g(0)=n=5. Then 16m+5=20 ⇒ 16m=15.
$m=\frac{15}{16}$, n=5.

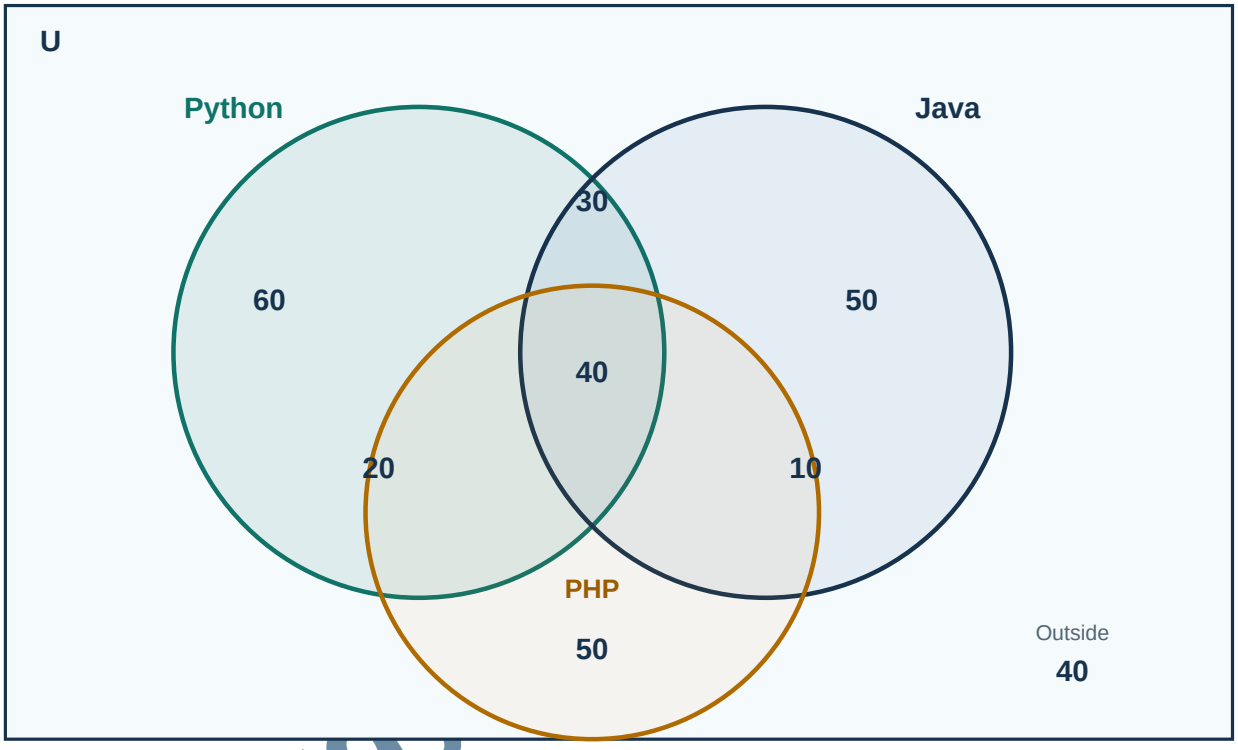
14. List the shopping-mall product sets and find A∪B∪C.
A={1,2,3,…,30}
B={31,32,33,…,55}
C={76,77,78,…,100}.
The three sets are disjoint, so concatenate their labels.
A∪B∪C={1,2,…,55,76,77,…,100}.

15. Of 180 students, 120 passed mathematics, 90 science and 60 both.
(a) Either = 120+90−60
= 150
(b) neither = 180−150
= 30.
(c) Science but not mathematics = 90−60 = 30
(d) failed science = 180−90
= 90.

16. Among 300 developers: Python 150, Java 130, PHP 120; pairwise overlaps 70, 60, 50; all three 40.

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At least one = 150+130+120−70−60−50+40
= 260; none
= 300−260
= 40.
Only Python = 150−(70−40)−(60−40)−40 = 60.
Only Java = 130−30−10−40 = 50; only PHP = 120−20−10−40 = 50.
Exactly one language = 60+50+50
= 160.
(a) 260
(b) 160
(c) 40
(d) 50.



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